$Step - 1: Read\ the\ data$

$Step - 2: Preprocess\ the\ data$

$Step - 3: Model\ development$

- $linear\ regression\ model$
- $y = b_o + b_1 * x_1 + b_2 * x_2 + \cdots . + b_n * x_n$
- $Goal: Find\ the\ suitable\ coeff\ in\ order\ to\ minimise\ the\ cost\ function$
- $OLS\ method$

$Step - 4: Model\ evaluation$

- MSE
- $RMSE$
- RSE
- $R_{squared}\ ====== > Over\ fitting\ - under\ fitting - Normal\ or\ Bias - varaince$
- $Adjusted\ R_{square}$
- AIC
- BIC

$Step - 5: Hyper\ parameter\ tuning$

$Step - 6: Model\ deployment$

- $Adjusted\ R_{square}$

$We\ already\ know\ that\ R_{square}$

- $R_{square} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{sum\ of\ squared\ residuals\ (SSR)}{Total\ sum\ of\ squares(SST)}$
- $Coefficient\ of\ determination$
- $Good\ ness\ of\ fit: How\ the\ model\ is\ fitting\ very\ well$
- $Ranges\ from\ 0\ to\ 1$
- $Explaniabity\ of\ the\ model:$
  $How\ much\ the\ input\ variables\ are\ explaining\ about\ output\ variable$

$Problem:$
$If\ you\ increase\ the\ columns\ R_{square}\ value\ incresaes$
$R_{Square}\ value\ never\ care\ about\ importance\ of\ features$
$we\ might\ misguided\ in\ order\ to\ get\ 100\%\ R_{square}, we\ will\ use\ all\ the\ features$
$This\ leads\ to\ model\ overfitting$

$In\ order\ to\ overcome\ this\ will\ go\ for\ Adjusted\ R_{Square}$

$Adjusted\ R_{Square}:$

- $It\ will\ adjust\ the\ R_{Square}\ \ based\ on\ features$
- $It\ will\ give\ accurate\ measurement\ of\ the\ model\ based\ on\ features$
- $whether\ the\ feature\ we\ are\ using\ to\ build\ to\ model\ is\ makes\ impact\ or\ not$

$Question$

$Im\ adding\ one$

$varaiable\ or$

$feature\ or$

$predictor\ or$

$independent\ variable\ \ \ is\ this\ really\ helpful\ to\ incresae\ the\ model\ performance$

$The\ answer\ given\ by\ Adjuster\ R_{Square}$

$$R_{Adj}\ =\ 1-\frac{\left[1-R_{square}\right]*(N-1)}{N-p-1}$$

$N = Total\ number\ of\ observations$

$p = predictors\ \ or\ \ independent\ variables$

$Case-1:\ \ \ p=0$

$$R_{Adj}\ =\ 1-\frac{\left[1-R_{square}\right]*(N-1)}{N-0-1}$$

$$R_{Adj}\ =\ 1-\frac{\left[1-R_{square}\right]*(N-1)}{N-1}$$

$$R_{Adj}\ =\ 1-1+R_{square}$$

$$R_{Adj}\ = R_{square}$$

$Conclusion:\ \ \ R_{Adj}\ \ maximum\ value = R_{square}$

$R_{Adj}\ \ minimum\ value = 0$

$Case - 2: \quad p = max$

$$R_{Adj} = 1 - \frac{\left[1 - R_{square}\right] * (N - 1)}{N - max - 1}$$

x_1 , x_2, x_3

$Ex - 1 \;\; y = b_o + b_1 * x_1 + b_2 * x_2$
$Assume \;\; R_{square} = 0.9 \;\; N = 10 \;\; P = 2 \quad R_{adj} = 0.871$

$$R_{Adj} = 1 - \frac{[1 - 0.9] * (N - 1)}{N - 2 - 1}$$

$Ex - 2:$

$N = 10 \quad p = 7 \quad Rsq = 0.92 \quad R_{adj} = 0.64$

$$R_{Adj} = 1 - \frac{[1 - 0.9] * (N - 1)}{N - 7 - 1} = 1 - \frac{[1 - 0.9] * (9)}{2} = 0.64$$

p	Rsq	R_{adj}	
2	0.9	0.87	
5			
7	0.95	0.7	

$Model - 2: \;\; y = b_o + b_1 * x_1 + b_2 * x_2 + b_3 * x_3$
$R_{square} = 0.91 \quad R_{adj} = 0.865$

$$R_{Adj} = 1 - \frac{[1 - 0.91] * (N - 1)}{N - 3 - 1}$$

$Model - 2.1: \;\; y = b_o + b_1 * x_1 + b_2 * x_2 + b_3 * x_3$
$R_{square} = 0.99 \quad R_{adj} = 0.985$

$$R_{Adj} = 1 - \frac{[1 - 0.99] * (N - 1)}{N - 3 - 1}$$

Conclusion:
In the model 2 when we add x_3 R_{adj} *significantly drop*
which means x_3 *variable should not use in model development*

Adjusted R_{Square} can use as feature selection method as forward selection or Backward selection process

feature selection method: < write your understanding here >

Backward selection method: < write your understanding here >

AIC: Akiaki information Creteria

$AIC: \quad 2P - 2\ln(L)$

$\ln = \log_e = 2.303 \log_{10}$

P = number of predictors or features

L = likelyhood estimation

$\ln(L)$ = log likeliyhood estimation

Normal scale vs Log scale

Conclusion

AIC is should low that is better model

We will talk about AIC for the complex model , means more predictors

If any model using more predictors we will use AIC method to conclude the model performance

$BIC: Baysian\ information\ Creteria$

$BIC: \quad p * ln(N) - 2\ln(L)$

$P = number\ of\ predictors\ \ or\ features$

$n = number\ of\ data\ points$

$L = likelyhood\ estimation$

$\ln(L) = \log likeliyhood\ estimation$

$Conclusion$

$BIC\ \ is\ \ should\ low\ that\ is\ better\ model$

$We\ will\ talk\ about\ BIC\ \ for\ the\ complex\ model\ , means\ more\ data\ points$

$If\ any\ model\ using\ \ more\ datapoints\ we\ will\ use\ \ \ BIC\ method\ to\ conclude\ the\ model$

$performance$